

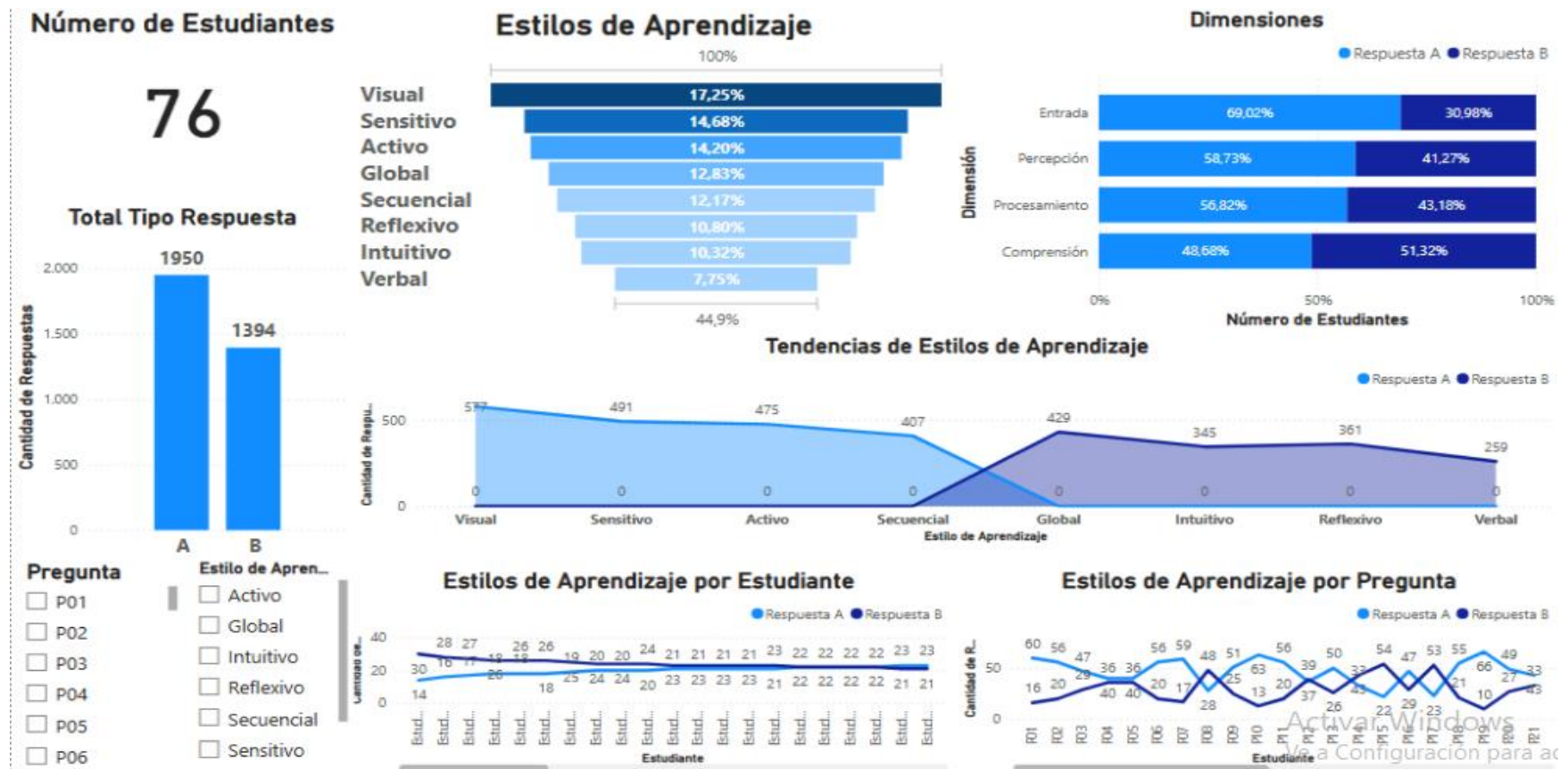
PREDICCIÓN DE ESTILOS DE APRENDIZAJE MEDIANTE INTELIGENCIA ARTIFICIAL

Learning Analytics
Educational Data Mining
Machine Learning
Inteligencia Artificial

Springer LNNS
Indexado en Scopus

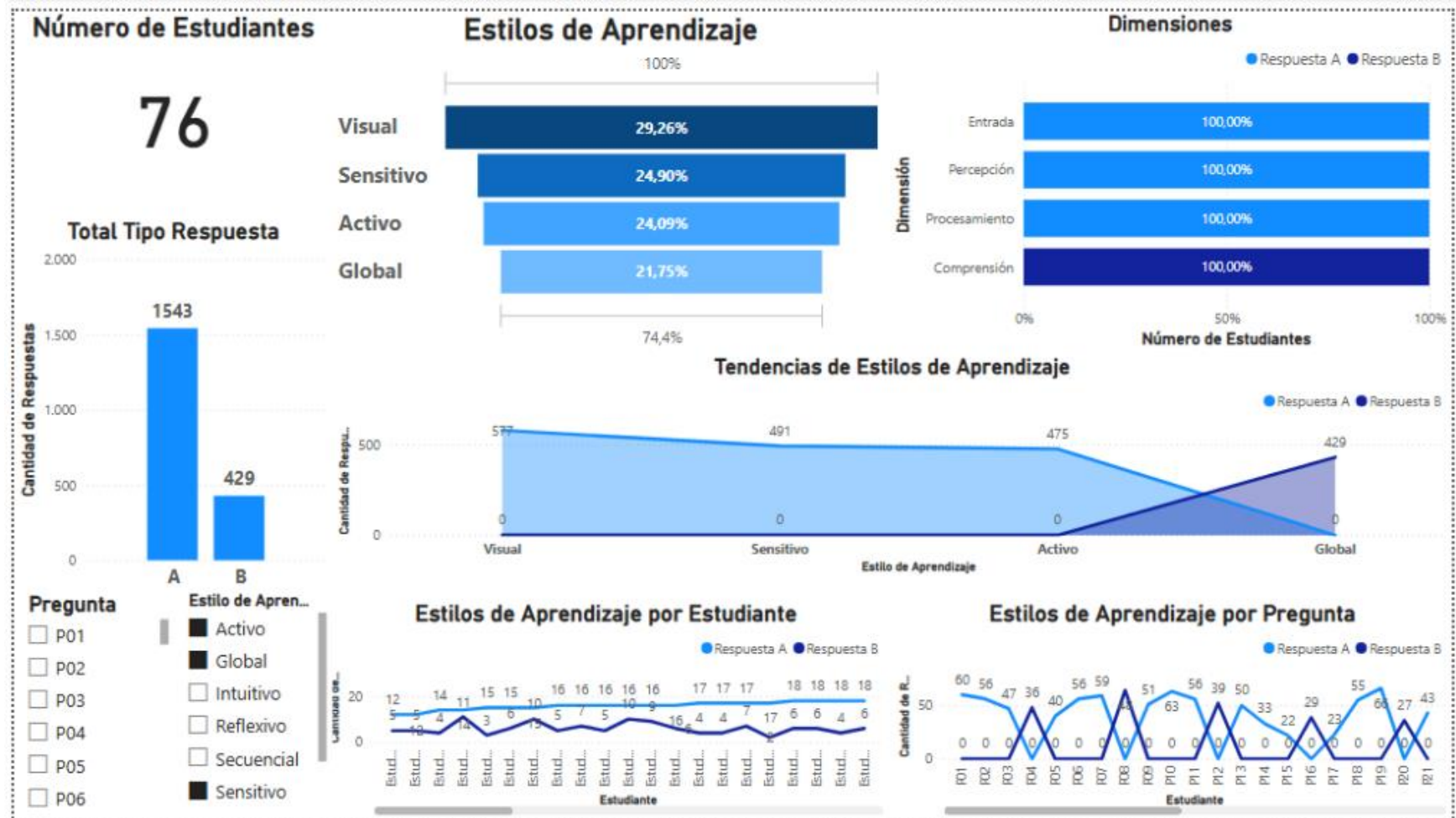
Problema de Investigación

- 76 estudiantes.
- 3500 respuestas.
- Modelo Felder y Silverman.
- 4 dimensiones de aprendizaje.

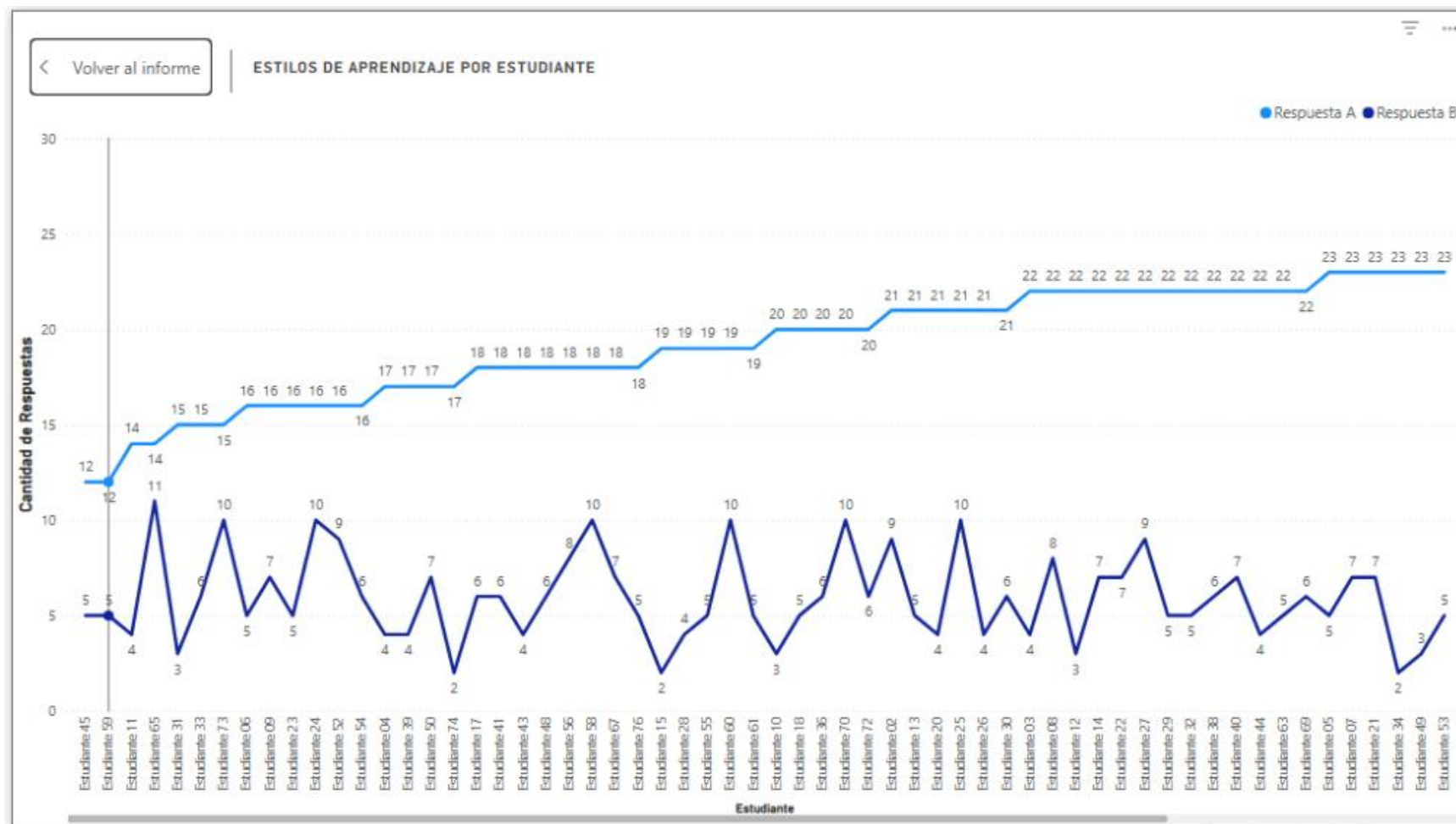


Resultados de Analítica de Datos

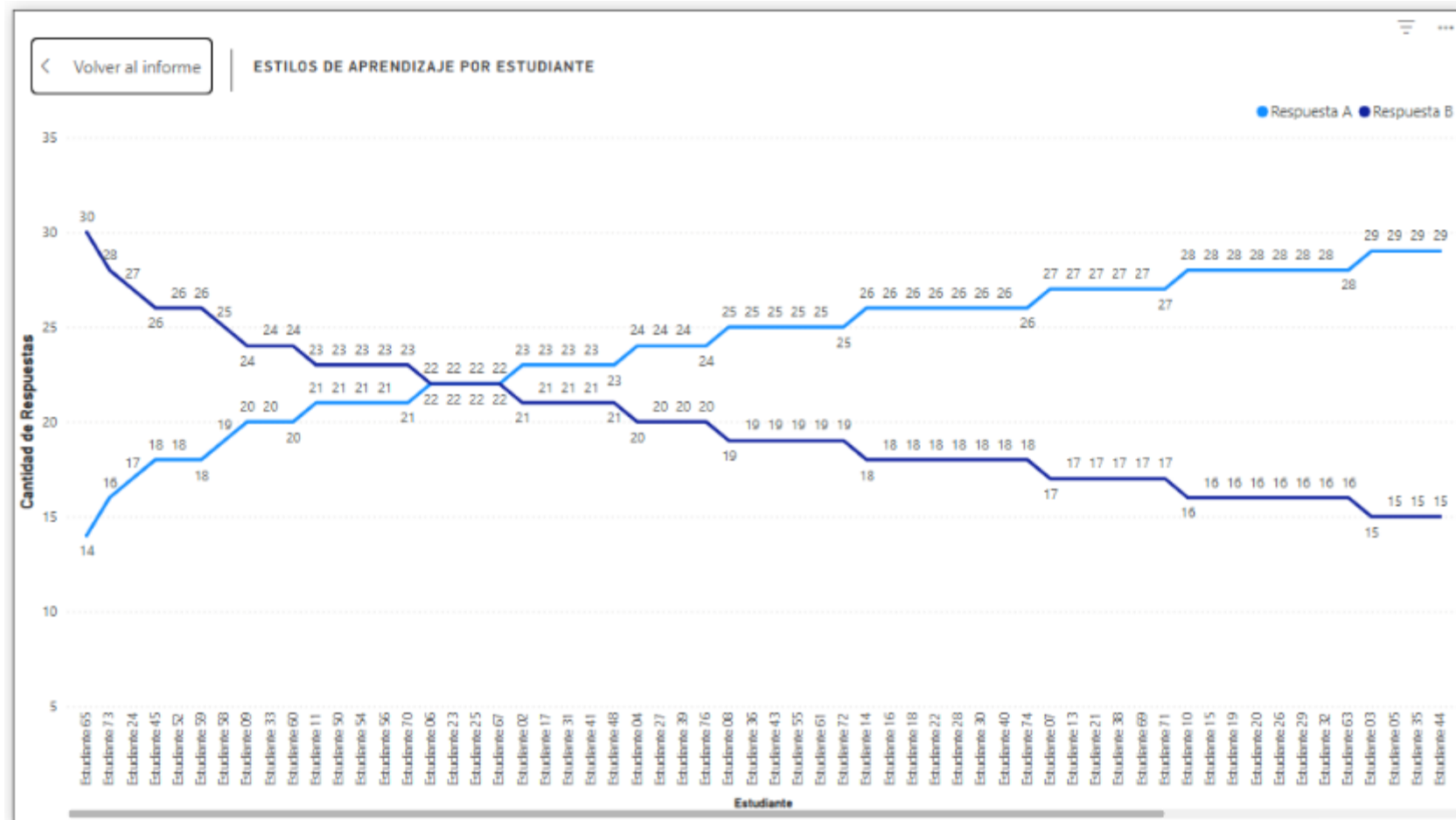
Dashboard de Estilos Predominantes



Dashboard de Tendencias Estilo Predominante



Dashboard de Tendencias Practicos vs Innovadores



Arquitectura de Inteligencia Artificial

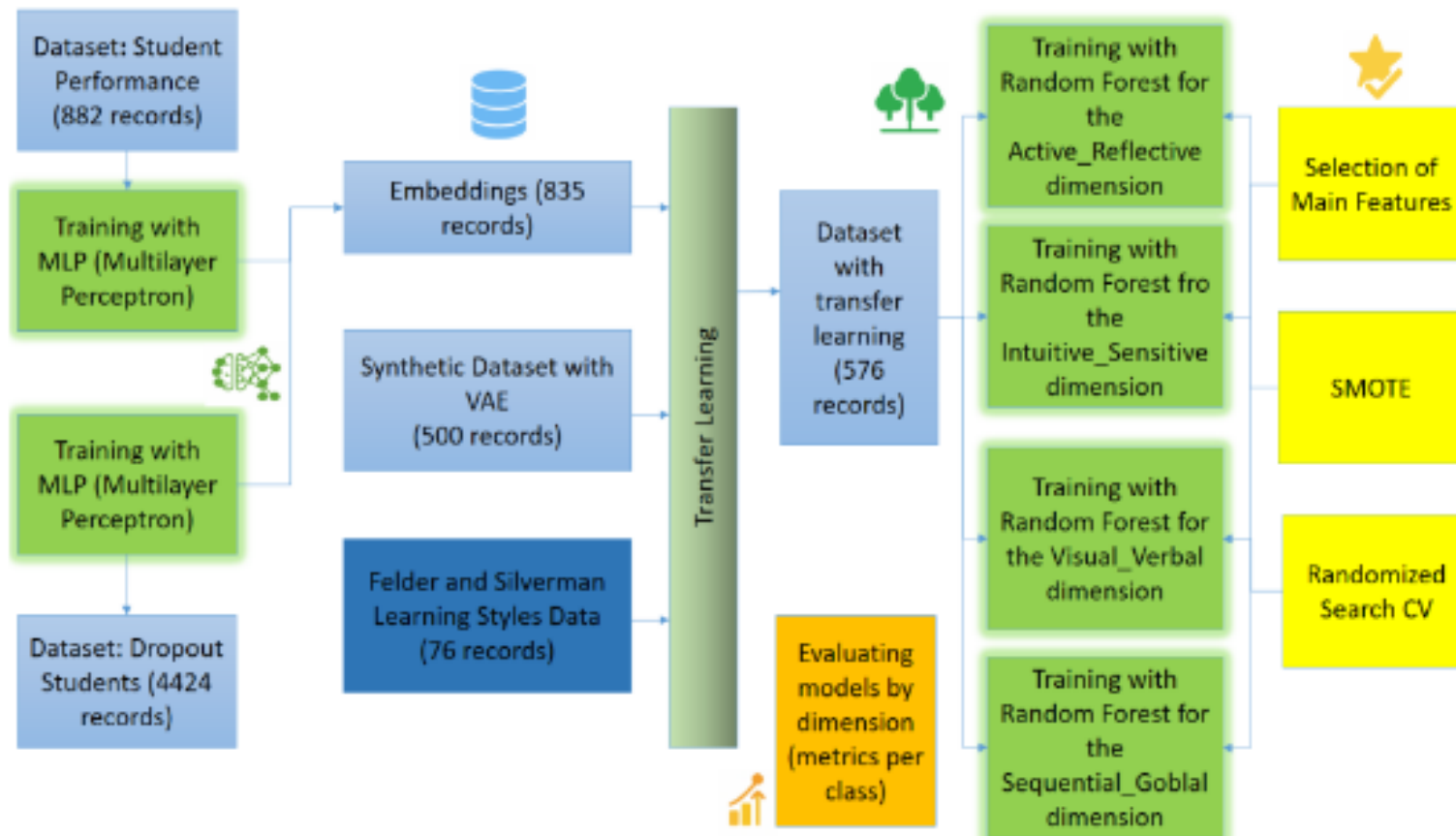


Fig. 3. AI Architecture for Learning Styles

Curvas de Predicción

Curvas ROC y Precision-Recall

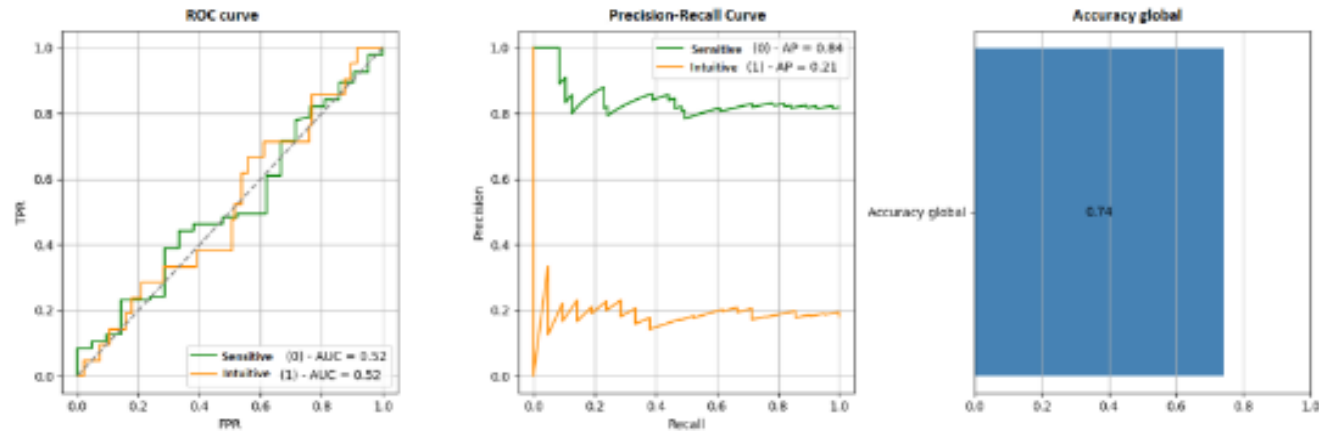


Fig. 4. Prediction Curves for the Perception Dimension (Sensing / Intuitive)

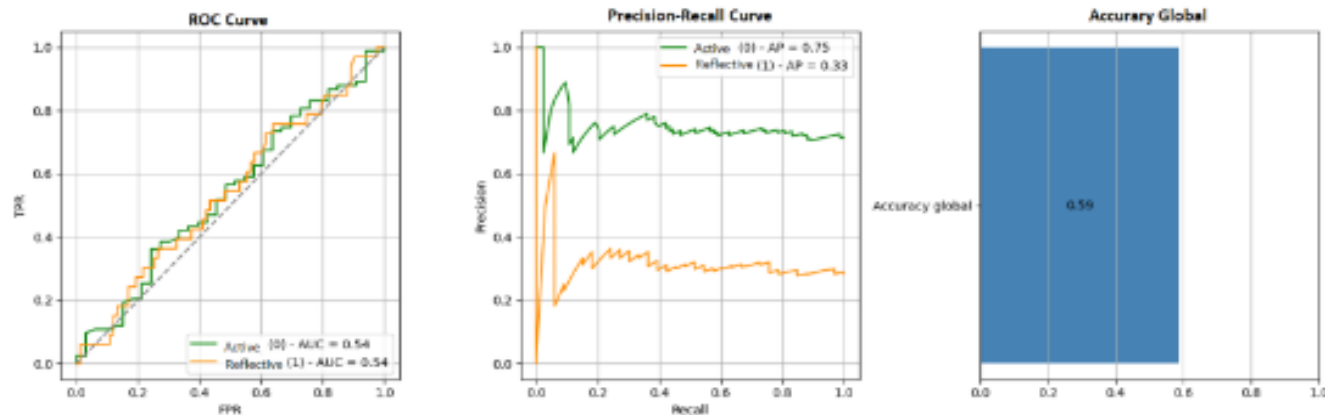


Fig. 6. Prediction Curves for the Processing Dimension (Active / Reflective)

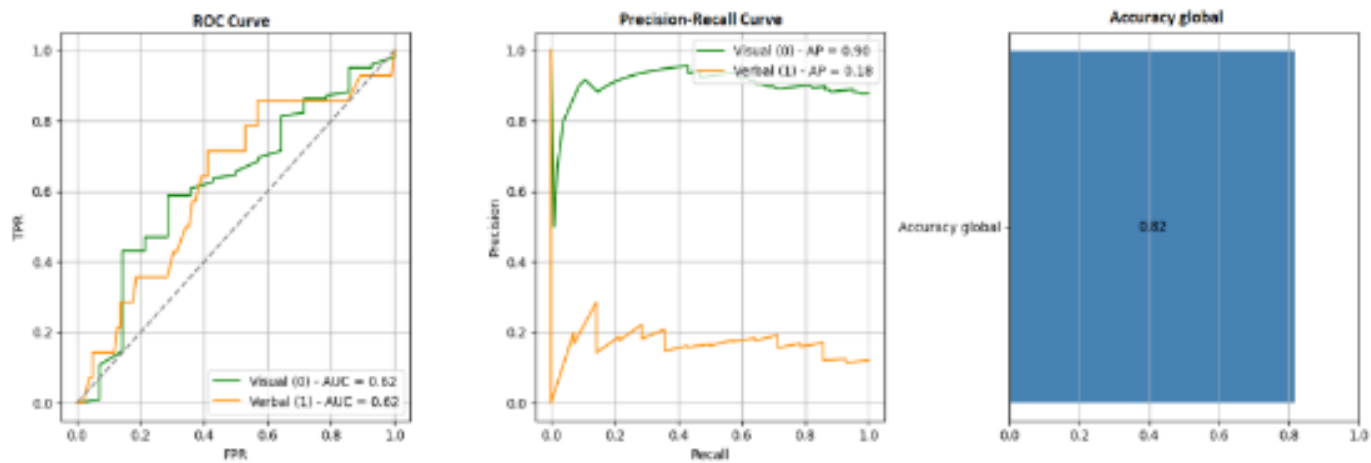


Fig. 8. Prediction Curves for the Representation Dimension (Visual / Verbal)

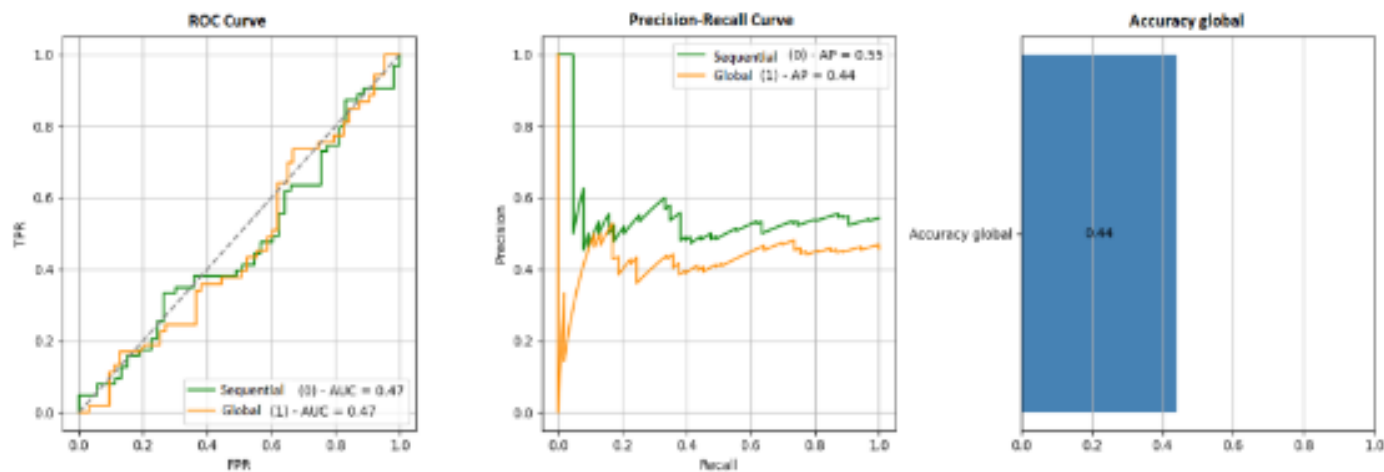


Fig. 10. Prediction Curves for the Understanding Dimension (Sequential / Global)

Matrices de Confusión y Métricas

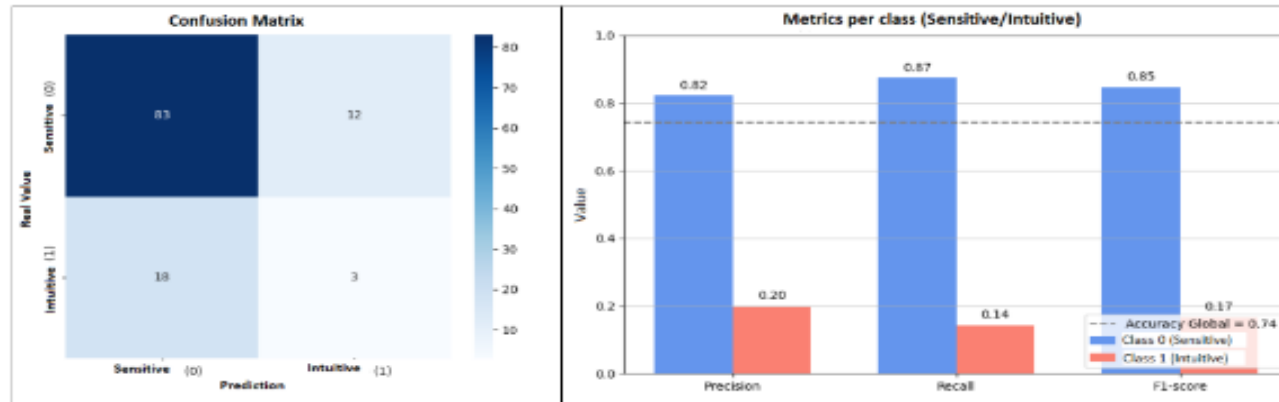


Fig. 5. Confusion Matrix and Prediction Metrics for the Perception Dimension

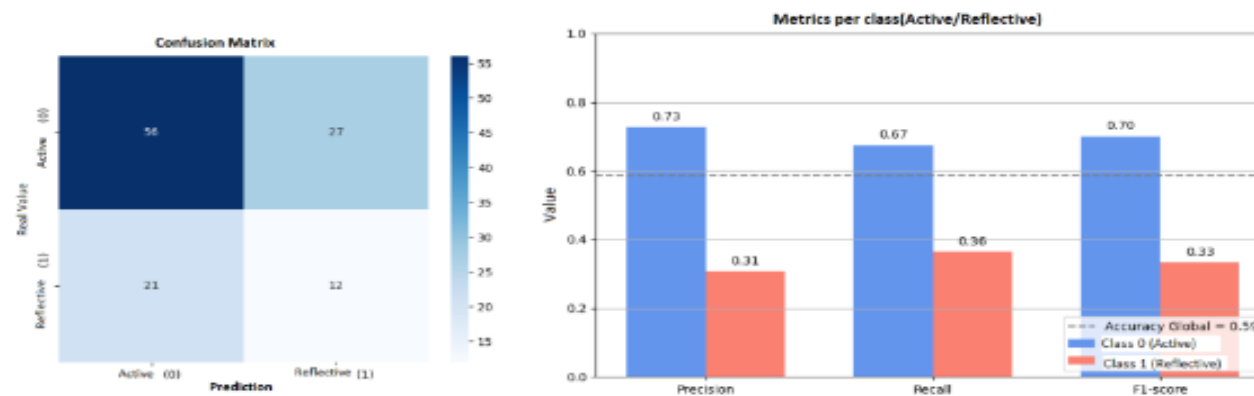


Fig. 7. Confusion Matrix and Prediction Metrics for the Processing Dimension

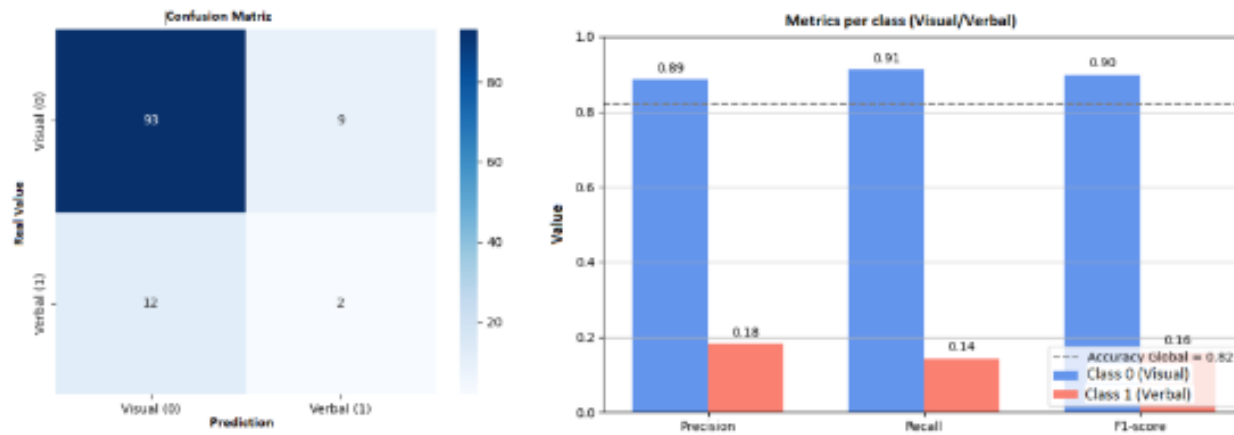


Fig. 9. Confusion Matrix and Prediction Metrics for the Representation Dimension

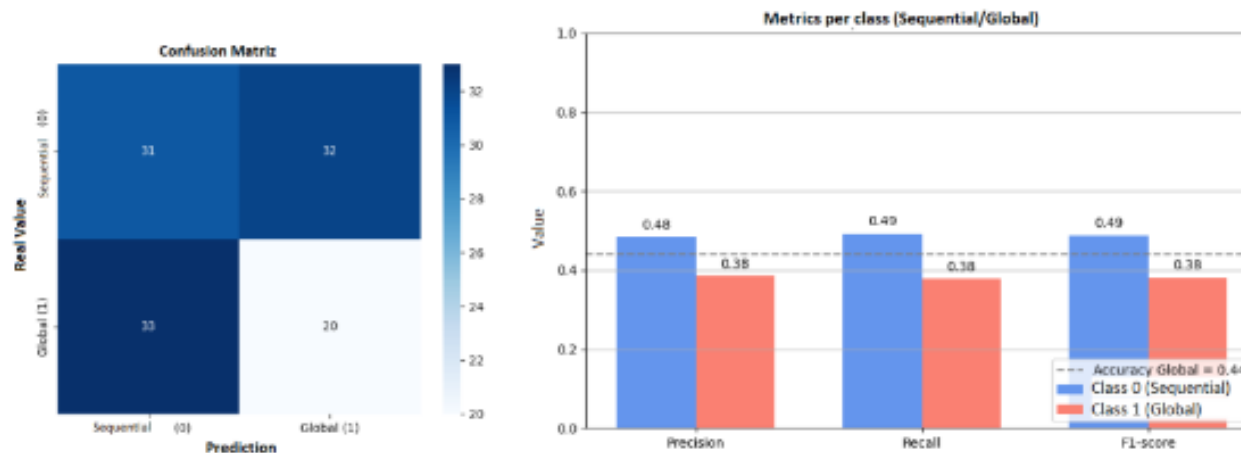


Fig. 11. Confusion Matrix and Prediction Metrics for the Understanding Dimension

Resultados de Inteligencia Artificial

- Sensitivo / Intuitivo => Accuracy: 0.74
- Activo / Reflexivo => Accuracy: 0.59
- Visual / Verbal => Accuracy: 0.82
- Secuencial / Global => Accuracy: 0.44

Impacto y Aplicaciones

- Learning Analytics
- Educational Data Mining
- Sistemas Adaptativos
- Inteligencia Artificial aplicada a Educación
- Personalización del Aprendizaje
- Predicción de Estilos de Aprendizaje
- Investigación Científica